## ENUNCIADO — PROBLEMA 1

**Numerical Analysis Preliminary Examination questions**
**August 2016**

<u>**Instructions**</u>: NO CALCULATORS are allowed. This examination contains **seven** problems, each worth 20 points. Do any **five** of the seven problems. Show your work. Clearly indicate which **five** are to be graded.

**PLEASE GRADE PROBLEMS: 1 2 3 4 5 6 7**

1. Calculate the following sum by hand in IEEE double precision computer arithmetic, using the Rounding to Nearest Rule:

$$(1 + (2^{-51} + 2^{-52} + 2^{-60})) - 1.$$

2. (a) Show how to calculate $\sqrt{5}$ using Newton's method.

   (b) Perform two iterations of Newton's method starting with an initial guess 2.

   (c) Prove that the Newton's method generates a sequence that converges to $\sqrt{5}$ with a quadratic rate.

3. (a) Find the solution to the following boundary value problem (BVP)

$$y'' = 2,\ 0 < t < 1,\ y(0) = 0,\ y(1) = 1.$$

   using a numerical method of your choice with 3 mesh points 0, 0.5 and 1.

   (b) Find the analytical solution to the BVP.

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4. (a) Write down the Lagrange interpolating polynomial $P(t)$ for the points $(0, 3), (1, 7), (2, 5)$.

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**SOLUCIÓN — PROBLEMA 1**

## Problema 1 — Aritmética IEEE de doble precisión

Se pide calcular a mano, con redondeo al más cercano ("round to nearest"):

$$(1 + (2^{-51} + 2^{-52} + 2^{-60})) - 1.$$

**Paso 1 — suma interna.** Los tres sumandos $2^{-51}, 2^{-52}, 2^{-60}$ tienen bits no nulos en las posiciones (tras el punto binario) $51, 52, 60$. La suma $s = 2^{-51} + 2^{-52} + 2^{-60}$ requiere, en su representación binaria normalizada (bit inicial en la posición 51), solo 10 bits de mantisa (de la posición 51 a la 60). Como el doble IEEE tiene 52 bits de mantisa (53 con el bit implícito), $s$ se representa **exactamente**, sin redondeo en este paso.

**Paso 2 — suma $1 + s$.** El número $1$ está en $[1, 2)$, así que un doble en ese rango solo puede resolver 52 bits de fracción **después del punto binario** (posiciones $1$ a $52$). El valor exacto de $1 + s$ tiene bits en las posiciones $51, 52$ (representables) y $60$ (fuera de rango: no representable). Hay que redondear.

La media unidad en el último lugar (ULP) en esta escala es $2^{-53}$ (la mitad de $2^{-52}$). Comparemos la contribución "perdida", $2^{-60}$, contra ese umbral:

$$2^{-60} = 2^{-53} \cdot 2^{-7} = \frac{2^{-53}}{128} \ll 2^{-53}.$$

Como $2^{-60}$ es muchísimo menor que medio ULP, el valor exacto $1 + 2^{-51} + 2^{-52} + 2^{-60}$ está mucho más cerca del número representable inferior, $1 + 2^{-51} + 2^{-52}$, que del superior, $1 + 2^{-51} + 2^{-52} + 2^{-52}$. Por la regla de redondeo al más cercano:

$$fl(1 + s) = 1 + 2^{-51} + 2^{-52} = 1 + 3 \cdot 2^{-52}$$

(este valor sí es exactamente representable: los bits $51$ y $52$ están dentro del rango de 52 bits de mantisa).

**Paso 3 — resta final.** $fl(1 + s) - 1$. Ambos operandos son exactamente representables y su diferencia, $2^{-51} + 2^{-52} = 3 \cdot 2^{-52}$, también lo es (la resta se realiza sin error adicional, por el Lema de Sterbenz o simplemente porque el resultado cae exactamente en la rejilla de dobles).

$$\boxed{(1 + (2^{-51} + 2^{-52} + 2^{-60})) - 1 = 2^{-51} + 2^{-52} = 3 \cdot 2^{-52} \approx 6.6613 \times 10^{-16}.}$$

**Verificación:** $2^{-52} \approx 2.22045 \times 10^{-16}$, así que $3 \times 2^{-52} \approx 6.6613 \times 10^{-16}$, y el término $2^{-60}\approx 8.67\times10^{-19}$ efectivamente desaparece por redondeo (queda absorbido, es menor que medio ULP en la escala de $1 + s$).

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1

## SOLUCIÓN — PROBLEMA 2

## Problema 2 — Newton para $\sqrt{5}$

### (a) Fórmula

Sea $f(x) = x^2 - 5$, $f'(x) = 2x$. Newton-Raphson:

$$x_{n+1} = x_n - \frac{x_n^2 - 5}{2x_n} = \frac{x_n^2 + 5}{2x_n} = \frac{1}{2}\left(x_n + \frac{5}{x_n}\right).$$

### (b) Dos iteraciones desde $x_0 = 2$

$$x_1 = \frac{1}{2}\left(2 + \frac{5}{2}\right) = \frac{1}{2} \cdot \frac{9}{2} = \frac{9}{4} = 2.25.$$

$$x_2 = \frac{1}{2}\left(\frac{9}{4} + \frac{5}{9/4}\right) = \frac{1}{2}\left(\frac{9}{4} + \frac{20}{9}\right) = \frac{1}{2} \cdot \frac{81 + 80}{36} = \frac{161}{72} \approx 2.236111.$$

**Verificación:** $\sqrt{5} = 2.2360679\ldots$; en efecto $x_1 = 2.25$ y $x_2 \approx 2.236111$ se acercan rápidamente (el error se cuadra en cada paso, como se muestra a continuación).

### (c) Convergencia cuadrática

Sea $e_n = x_n - \sqrt{5}$. Entonces

$$x_{n+1} - \sqrt{5} = \frac{x_n^2 + 5}{2x_n} - \sqrt{5} = \frac{x_n^2 - 2\sqrt{5}\,x_n + 5}{2x_n} = \frac{(x_n - \sqrt{5})^2}{2x_n} = \frac{e_n^2}{2x_n}.$$

Además, por la desigualdad AM-GM, para $x_0 > 0$ y todo $n \geq 0$:

$$x_{n+1} = \frac{1}{2}\left(x_n + \frac{5}{x_n}\right) \geq \sqrt{x_n \cdot \frac{5}{x_n}} = \sqrt{5},$$

así que $x_n \geq \sqrt{5} > 0$ para todo $n \geq 1$ (los iterados están acotados lejos de cero). Por lo tanto:

$$0 \leq e_{n+1} = \frac{e_n^2}{2x_n} \leq \frac{e_n^2}{2\sqrt{5}} \qquad (n \geq 1),$$

lo que muestra $e_{n+1} = O(e_n^2)$: **convergencia cuadrática**, con constante asintótica $C = \lim_{n\to\infty} e_{n+1}/e_n^2 = 1/(2\sqrt{5}) \approx 0.2236$. ■

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## SOLUCIÓN — PROBLEMA 3

## Problema 3 — BVP $y'' = 2$, $y(0) = 0$, $y(1) = 1$, malla $\{0, 0.5, 1\}$

### (a) Solución numérica (diferencias finitas centradas, $h = 0.5$)

En el único nodo interior $x_1 = 0.5$:

$$\frac{y_0 - 2y_1 + y_2}{h^2} = 2 \implies \frac{0 - 2y_1 + 1}{0.25} = 2 \implies 1 - 2y_1 = 0.5 \implies y_1 = 0.25.$$

### (b) Solución analítica

$y'' = 2 \Rightarrow y(x) = x^2 + Cx + D$. Con $y(0) = 0$: $D = 0$. Con $y(1) = 1$: $1+C=1 \Rightarrow C=0 $.Luego$ y(x)=x^2, $yenparticular$ y(0.5)=0.25$.

### (c) Error máximo

En los tres nodos de malla ($x = 0, 0.5, 1$) la solución numérica coincide **exactamente** con la analítica: $y_0 = 0 = 0^2$, $y_1 = 0.25 = 0.5^2$, $y_2 = 1 = 1^2$. El error nodal máximo es, por lo tanto, **cero**.

Esto no es casualidad: la fórmula de diferencia centrada de segundo orden para $u''$ tiene error de truncamiento $-\frac{h^2}{12}y^{(4)}(\xi)$, y aquí $y(x) = x^2$ es un polinomio de grado $2$, así que $y^{(4)} \equiv 0$: el esquema es exacto en los nodos para esta solución particular.

*(Si en cambio se compara la parábola exacta $y = x^2$ con el interpolante lineal a trozos de los tres valores nodales — es decir, el error entre nodos — el error máximo es $1/16 = 0.0625$, alcanzado en $x = 1/4$ y $x = 3/4$: en $[0, 0.5]$ el interpolante es $\ell(x) = 0.5x$, y $d(x) = \ell(x) - x^2$ tiene máximo en $x = 0.25$ con $d(0.25) = 0.125 - 0.0625 = 0.0625$; por simetría, lo mismo ocurre en $[0.5, 1]$. Pero el error **nodal**, que es lo que típicamente se pide, es 0.)* ■

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$$\int_{-1}^{1} f(x)\,dx = w_1 f(-a) + w_2 f(0) + w_3 f(a)$$

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$$\frac{dy}{dt} = a + by(t) + c\cos y(t), \qquad 0 \le t \le 1$$

   where $y(0) = 1$ and $a, b, c > 0$ are constants. Let us suppose that the solution satisfies $\max_{0\le t\le 1} |y''(t)| = M < \infty$. Consider the approximation $y_{k+1} = y_k + (a + by_k + c\cos(y_k))h$ for $k = 0, 1, 2, \ldots, N-1$ and let $y_0 = y(0)$ and $h = \dfrac{1}{N}$. Prove that

$$|y(1) - y_N| \le \frac{Mhe^{b+c}}{2(b+c)}.$$

2

## SOLUCIÓN — PROBLEMA 4

## Problema 4 — Interpolación de Lagrange, sistema de Vandermonde, LU y Householder

Datos: $(0,3),(1,7),(2,5)$.

### (a) Polinomio de Lagrange

$$P(t)=3\cdot\frac{(t-1)(t-2)}{(0-1)(0-2)}+7\cdot\frac{t(t-2)}{(1-0)(1-2)}+5\cdot\frac{t(t-1)}{(2-0)(2-1)}=\frac{3}{2}(t-1)(t-2)-7t(t-2)+\frac{5}{2}t(t-1).$$

Expandiendo: $\frac32(t^2-3t+2)-7(t^2-2t)+\frac52(t^2-t) =(1.5-7+2.5)t^2+(-4.5+14-2.5)t+3 = -3t^2+7t+3.$

$$\boxed{P(t)=-3t^2+7t+3.}$$

**Verificación:** $P(0)=3$ ✓, $P(1)=-3+7+3=7$ ✓, $P(2)=-12+14+3=5$ ✓.

### (b) Sistema de Vandermonde

Con $P(t)=c_0+c_1t+c_2t^2$, imponiendo $P(t_i)=y_i$:

$$A=\begin{pmatrix}1&0&0\\1&1&1\\1&2&4\end{pmatrix},\qquad b=\begin{pmatrix}3\\7\\5\end{pmatrix},\qquad Ax=b,\ x=(c_0,c_1,c_2)^T.$$

(En efecto $x=(3,7,-3)^T$ resuelve $Ax=b$, coincidiendo con (a).)

### (c) Factorización $LU$ con pivoteo parcial

**Columna 1** (entradas $1,1,1$): empate; no hay intercambio, se usa la fila 1 como pivote. Multiplicadores $m_{21}=m_{31}=1$. Restando la fila 1 (escalada) de las filas 2 y 3:

$$\begin{pmatrix}1&0&0\\0&1&1\\0&2&4\end{pmatrix}.$$

**Columna 2** (submatriz filas 2–3: entradas $1,2$): el pivote parcial exige el de mayor módulo, 2 (fila 3); **se intercambian las filas 2 y 3**:

$$\begin{pmatrix}1&0&0\\0&2&4\\0&1&1\end{pmatrix}.$$

Multiplicador $m_{32}=1/2$: fila 3 ← fila 3 $-\frac{1}{2}$ ·fila 2: $(0,1-1,1-2)=(0,0,-1)$.

$$U = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 4 \\ 0 & 0 & -1 \end{pmatrix}, \quad L = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & \frac{1}{2} & 1 \end{pmatrix}, \quad P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \text{ (intercambia filas 2 y 3).}$$

**Verificación ($PA = LU$):**

$$PA = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 2 & 4 \\ 1 & 1 & 1 \end{pmatrix}, \qquad LU = \begin{pmatrix} 1 & 0 & 0 \\ 1 \cdot 1 & 1 \cdot 2 & 1 \cdot 4 \\ 1 \cdot 1 + \frac{1}{2} \cdot 0 & \frac{1}{2} \cdot 2 & \frac{1}{2} \cdot 4 - 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 2 & 4 \\ 1 & 1 & 1 \end{pmatrix}.$$

Coinciden. ■

### (d) Transformación de Householder para eliminar la columna 1

Sea $a = (1, 1, 1)^T$ (columna 1 de $A$), $\|a\|_2 = \sqrt{3}$. Se elige $\alpha = -\text{sign}(a_1)\|a\|_2 = -\sqrt{3}$ (signo opuesto a $a_1$, para evitar cancelación catastrófica). El vector de Householder:

$$v = a - \alpha e_1 = (1 + \sqrt{3},\ 1,\ 1)^T, \qquad v^T v = (1 + \sqrt{3})^2 + 1 + 1 = 6 + 2\sqrt{3}.$$

La transformación:

$$H = I - \frac{2}{v^T v} v v^T = I - \frac{1}{3 + \sqrt{3}} v v^T,$$

que satisface $Ha = \alpha e_1 = (-\sqrt{3}, 0, 0)^T$, eliminando (llevando a cero) las entradas $2$ y $3$ de la primera columna, como se requiere. $H$ es simétrica y ortogonal ($H^T H = I$), como toda reflexión de Householder.

## ENUNCIADO — PROBLEMA 5

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   where $y(0) = 1$ and $a, b, c > 0$ are constants. Let us suppose that the solution satisfies $\max\limits_{0\le t\le 1} |y''(t)| = M < \infty$. Consider the approximation $y_{k+1} = y_k + (a + by_k + c\cos(y_k))h$ for $k = 0, 1, 2, \ldots, N-1$ and let $y_0 = y(0)$ and $h = \dfrac{1}{N}$. Prove that

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2

## SOLUCIÓN — PROBLEMA 5

## Problema 5 — Algoritmo QR para valores propios

Recurrencia: $A_0 = A$; en cada paso, $A_{k-1} = Q_k R_k$ (factorización QR), $A_k := R_k Q_k$.

### (a) Todas las $A_k$ tienen los mismos valores propios

De $A_{k-1} = Q_k R_k$ se sigue $R_k = Q_k^{-1} A_{k-1} = Q_k^T A_{k-1}$ (pues $Q_k$ es ortogonal). Sustituyendo en $A_k = R_k Q_k$:

$$A_k = Q_k^T A_{k-1} Q_k,$$

es decir, $A_k$ es **ortogonalmente semejante** a $A_{k-1}$. Como matrices semejantes tienen el mismo polinomio característico (y por tanto los mismos valores propios: si $A_{k-1}v \; = \lambda v$, entonces $A_k(Q_k^Tv)=Q_k^TA_{k-1}v =\lambda Q_k^Tv$), por inducción

$$A_k = (Q_1 \cdots Q_k)^T \, A_0 \, (Q_1 \cdots Q_k)$$

es semejante a $A_0 = A$ para todo $k \geq 0$. Luego todas las $A_k$ comparten el mismo espectro que $A$. ■

### (b) Determinación de vectores propios

Sea $Q^{(m)} := Q_1 Q_2 \cdots Q_m$ (matriz ortogonal, producto acumulado de los factores $Q$ de cada paso). De la parte (a), $A_m = (Q^{(m)})^T A_0 Q^{(m)} = \Lambda$ (diagonal, por hipótesis). Multiplicando por $Q^{(m)}$ a la izquierda:

$$A_0 \, Q^{(m)} = Q^{(m)} \Lambda.$$

Escribiendo esta igualdad columna por columna, si $q_i$ es la $i$-ésima columna de $Q^{(m)}$ y $\lambda_i = \Lambda_{ii}$:

$$A_0 \, q_i = \lambda_i \, q_i,$$

es decir, **las columnas de $Q^{(m)} = Q_1 Q_2 \cdots Q_m$ son los vectores propios de $A$**, correspondientes a los valores propios en la diagonal de $\Lambda = A_m$. (Como $A$ es simétrica con valores propios distintos, estos vectores propios son automáticamente ortonormales, consistente con que $Q^{(m)}$ es ortogonal.) En la práctica: se acumula el producto de los factores $Q_k$ de cada iteración; ese producto converge a la matriz de vectores propios. ■

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## SOLUCIÓN — PROBLEMA 6

## Problema 6 — Cuadratura de 3 puntos simétrica (Gauss-Legendre)

$$\int_{-1}^{1} f(x)\,dx = w_1 f(-a) + w_2 f(0) + w_3 f(a).$$

### (a) Coeficientes de máxima exactitud

Por simetría, si se impone exactitud para $f = x$: \$-aw_1+aw_3=0\Rightarrow w_1=w_3 $(asumiendo a\ne0); entonces$ f=x^3\$ se satisface automáticamente ($a^3(w_3 - w_1) = 0 = \int_{-1}^{1} x^3 dx$ ), sin dar información nueva. Quedan tres incógnitas efectivas $w_1(=w_3), w_2, a$, determinadas por exactitud en $f = 1, x^2, x^4$:

$$f = 1: \quad 2 = 2w_1 + w_2.$$

$$f = x^2: \quad \frac{2}{3} = 2w_1 a^2.$$

$$f = x^4: \quad \frac{2}{5} = 2w_1 a^4.$$

Dividiendo la tercera entre la segunda: $a^2 = \dfrac{2/5}{2/3} = \dfrac{3}{5}$, así $a = \sqrt{3/5}$. De $2w_1 a^2 = 2/3$: $w_1 = \dfrac{1/3}{a^2} = \dfrac{1/3}{3/5} = \dfrac{5}{9}$. De la primera: $w_2 = 2 - 2w_1 = 2 - \dfrac{10}{9} = \dfrac{8}{9}$.

$$\boxed{a = \sqrt{\tfrac{3}{5}}, \quad w_1 = w_3 = \tfrac{5}{9}, \quad w_2 = \tfrac{8}{9}.}$$

Esta es exactamente la **regla de cuadratura gaussiana de 3 puntos (Gauss-Legendre)**, exacta para polinomios hasta grado $5$ (verificación: $f = x^6$ da $2w_1 a^6 = 2 \cdot \frac{5}{9} \cdot (3/5)^3 = 0.24 \neq \frac{2}{7} \approx 0.2857$, así que grado $6$ ya falla — el grado máximo de exactitud es $2n - 1 = 5$ con $n = 3$ nodos, como predice la teoría de cuadratura gaussiana).

### (b) Aplicación a $\int_0^1 \frac{1}{x+1} dx$

Cambio de variable $x = \frac{t+1}{2}$ ($t \in [-1, 1] \to x \in [0, 1]$), $dx = dt/2$:

$$\int_0^1 \frac{dx}{x+1} = \int_{-1}^{1} \frac{1}{\frac{t+1}{2} + 1} \cdot \frac{dt}{2} = \int_{-1}^{1} \frac{dt}{t+3} =: \int_{-1}^{1} g(t)\,dt.$$

Aplicando la regla con $a = \sqrt{0.6} \approx 0.774597$:

$$g(-a) = \frac{1}{3-a} \approx \frac{1}{2.225403} \approx 0.449339, \quad g(0) = \frac{1}{3} \approx 0.333333, \quad g(a) = \frac{1}{3+a} \approx \frac{1}{3.774597} \approx 0.264928.$$

$$\int_{-1}^{1} g\,dt \approx \frac{5}{9}(0.449339 + 0.264928) + \frac{8}{9}(0.333333) \approx 0.555556 \times 0.714267 + 0.888889 \times 0.333333 \approx 0.396815 + 0.296296 = 0.693111.$$

**Verificación:** el valor exacto es $\int_0^1 \frac{dx}{x+1} = \ln 2 = 0.693147\ldots$ El error de la cuadratura, $\approx 3.6 \times 10^{-5}$, es minúsculo, consistente con la exactitud de grado 5 de la regla aplicada a una función suave no polinómica. ■

## ENUNCIADO — PROBLEMA 7

5. Consider the QR Iteration algorithm for a matrix $A = A_0$

   for k=1:m

   Compute the QR factorization: $Q_k R_k = A_{k-1}$

   Set $A_k = R_k Q_k$

   end

   (a) Show that the matrices $A_k$ all have the same eigenvalues.

   (b) Assume that $A$ is a real symmetric matrix with distinct eigenvalues and that $A_m$ is equal to a diagonal matrix $\Lambda$. Describe how to determine the eigenvectors of $A$ from the algorithm.

6. A quadrature rule is given by:

$$\int_{-1}^{1} f(x)\,dx = w_1 f(-a) + w_2 f(0) + w_3 f(a)$$

   (a) Find $w_1, w_2, w_3, a$ that will make this quadrature rule exact for all polynomials of as high a degree as possible.

   (b) Use this quadrature rule to explain how to numerically compute $\displaystyle\int_0^1 \frac{1}{x+1}\,dx$.

7. Consider the initial value problem

$$\frac{dy}{dt} = a + by(t) + c\cos y(t), \qquad 0 \le t \le 1$$

   where $y(0) = 1$ and $a, b, c > 0$ are constants. Let us suppose that the solution satisfies $\max\limits_{0\le t\le 1} |y''(t)| = M < \infty$. Consider the approximation $y_{k+1} = y_k + (a + by_k + c\cos(y_k))h$ for $k = 0, 1, 2, \ldots, N-1$ and let $y_0 = y(0)$ and $h = \dfrac{1}{N}$. Prove that

$$|y(1) - y_N| \le \frac{Mhe^{b+c}}{2(b+c)}.$$

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## SOLUCIÓN — PROBLEMA 7

## Problema 7 — Cota de error del Euler explícito no lineal

$y' = a + by + c\cos y$, $y(0) = 1$, $a, b, c > 0$, $\max_{[0,1]} |y''| = M < \infty$. Euler: $y_{k+1} = y_k + h\Phi(y_k)$, $\Phi(y) = a + by + c\cos y$, $h = 1/N$.

**Constante de Lipschitz de $\Phi$:** $\Phi'(y) = b - c\sin y$, así que $|\Phi'(y)| \le b + c$ (pues $|\sin y| \le 1$ y $b, c > 0$). Por el teorema del valor medio, $\Phi$ es Lipschitz en $y$ con constante $L = b + c$.

**Error de truncamiento local:** como $\Phi = f$ (Euler explícito estándar), por Taylor con residuo de Lagrange:

$$y(t_{k+1}) = y(t_k) + hy'(t_k) + \frac{h^2}{2}y''(\xi_k) = y(t_k) + h\Phi(t_k, y(t_k)) + \frac{h^2}{2}y''(\xi_k),$$

así que $\tau_k = \frac{h^2}{2}y''(\xi_k)$, con $|\tau_k| \le \frac{M}{2}h^2$.

**Recursión del error global** ($e_k = y_k - y(t_k)$, $e_0 = 0$), igual que en el Problema 6 del examen de enero de 2016:

$$|e_{k+1}| \le (1 + hL)|e_k| + \frac{M}{2}h^2,$$

y por inducción (idéntica a la del problema análogo):

$$|e_k| \le \frac{(M/2)h}{L}\left[(1 + hL)^k - 1\right] = \frac{Mh}{2L}\left[(1 + hL)^k - 1\right].$$

En $k = N$, $Nh = 1$, y usando $1 + x \le e^x$: $(1 + hL)^N \le e^{hLN} = e^L = e^{b+c}$. Por lo tanto:

$$|y(1) - y_N| = |e_N| \le \frac{Mh}{2(b + c)}\left(e^{b+c} - 1\right) \le \frac{Mh}{2(b + c)}\, e^{b+c},$$

pues $e^{b+c} - 1 < e^{b+c}$ trivialmente. Esto establece la cota pedida (de hecho una cota ligeramente más fuerte):

$$\boxed{|y(1) - y_N| \le \frac{Mhe^{b+c}}{2(b + c)}.} \blacksquare$$